

The source code of this project is available for free download at source.efymag.com

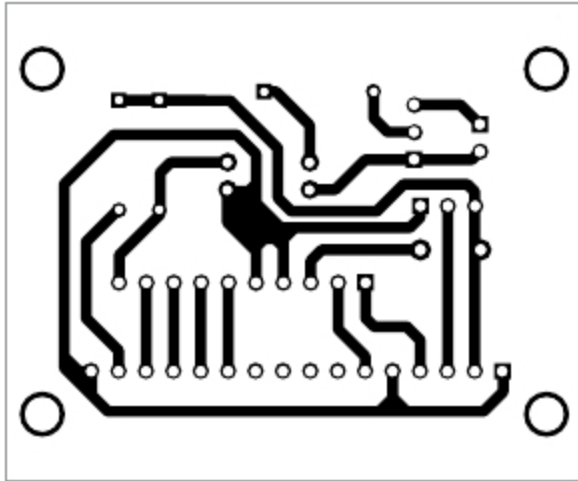


Fig. 3: Actual-size PCB layout of Arduino-based tachometer

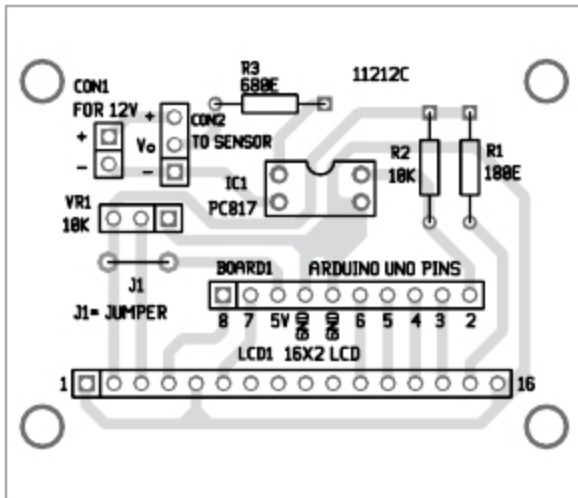


Fig. 4: Components layout for the PCB

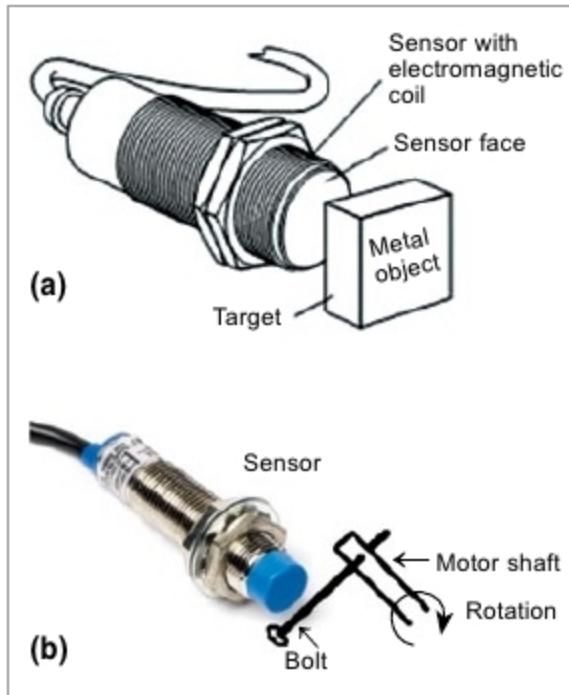


Fig. 5: Sensor arrangement

across CON1 for the sensor to operate. Place the proximity sensor near the object whose rpm is to be measured, as shown in Fig. 5. When the metal target comes near the proximity sensor, the inbuilt LED on the rear side of sensor blinks. This indicates that the sensor is sensing the metal target (object).

RPM measurement. You cannot

read the rpm directly because of resolution. Resolution is based on the number of metal target(s). The rpm can be measured using the following relationship:

$$\text{rpm} = [\text{interrupt count per second} \times 60(\text{secs per min})] / \text{number of metal targets}$$

If metal target is 1, resolution will be 60rpm; if metal target is 10, resolution will be 6rpm; if metal target is 60, resolution will be 1rpm, and so on. One metal target was used during testing at EFY Labs. As metal target increases, resolution also increases. **EFY**



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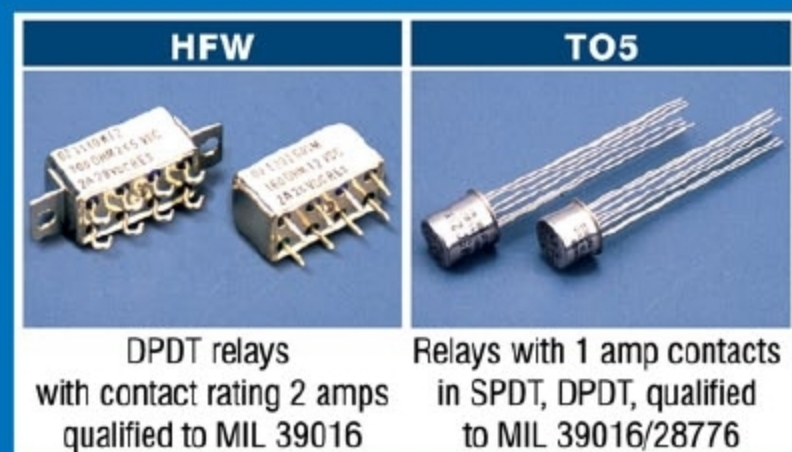
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